

Individual Contribution Report

Bipindra Rai

1.0 Introduction

SterLearning is a project developed in response to the problem statement proposed by Dominic Marcus Finance Consulting [6]. This initiative aimed to give young people the tools they need to understand finances and manage budgets, a critical area often overlooked in formal education. As a full-stack developer, my responsibilities encompassed both backend and frontend tasks. I ensured the application was both cohesive and functional, tailored to engage and educate the youth.

2.0 Contribution to the Group's Project

As a full-stack developer in the COM2027 software engineering project, my contributions spanned both backend and frontend developments, addressing comprehensive needs from system architecture to user interface design.

In this section I will list my most notable contributions:

2.1 Backend Development

- **Database Seeding and Content Development:** Developed the seed.py for testing [7] and developed seed.py[3] script for populating our database with data, efficiently loading structured data from JSON files into our Django models. This automation facilitated the development and testing environments significantly.
- **Quiz Content Creation:** Researched, Created and structured JSON files for six quizzes [3], containing the information for the quizzes on taxes and budgeting. Which was essential for the app to have quizzes to display.

2.2 Frontend Development

- **Header.html (without Bootstrap):** Developed a custom sidebar without relying on Bootstrap[4]. This initiative allowed for a more customised user experience.
- **Template and Footer Implementation:** Created the base.html template to ensure consistency across the application's main pages. Also designed and implemented the footer.html, which included custom JavaScript for dynamic UI interactions, enhancing the application's overall aesthetic and functionality. Seen in the git reference [6].
- **Research implementation:** Through my API research [2], we decided on the integration of Boxicons [5], which enhanced the application's user interface with intuitive iconography.

2.3 Documentation

- **For the PDD:** I was tasked with writing: 8.0 Risk Analysis (10%) [9] including 9.8 Risk Analysis table [10], 5.2.5 Bipin [11]
- **For the FAR:** I was tasked with writing: 4.6 Bipin [12], 3.0 Test Suites [13], 3.4 Delivery [14] , 3.3 Evaluation [15].
I was tasked with Editing for: 2.1 Front-End Design [16] , 3.0 Team Evaluation [17], 1.0 How to run the application [18].

3.0 Personal Gains from the Project

In my role as a full-stack developer, I achieved significant progress in my goal of enhancing my Git skills for version control. I advanced my ability to effectively manage branches and resolve merge conflicts, which enhanced our project's workflow.

Working closely with the test lead when doing research and implementing tests, I also deepened my understanding of software testing, contributing to both the design and execution of test cases. Although it was not originally one of my goals, I seized the opportunity

to expand my technical skills in JavaScript. I successfully developed the side header/navbar for our application (referenced in header.html [6]), this task diverged from our typical use of Bootstrap. This allowed me to employ more flexible and responsive frontend development techniques. A notable challenge was creating frontend HTML independently of backend constraints, which further honed my skills in building robust and scalable user interfaces.

4.0 What Went Well

Throughout the development of Sterlearning, our team maintained a conflict-free environment, which fostered open and effective communication. This understanding among the team members was crucial in smooth collaboration and quick resolution of issues that appeared. We had regular meetings and they were documented (meeting minutes, as seen in [1]). The meetings were weekly to ensure all team members were up to date with the project's progress and possible immediate actions. The meeting minutes (referenced in [1] & [2]) provided a reliable record, to help us stay organised and track our decisions and changes throughout the project's lifecycle. To effectively replicate this success in our future projects we must foster an environment of open communication and regular feedback, this will allow us to keep a supportive team environment, where obstacles are tackled with a collaborative approach. Not only does this accelerate problem solving but also enhance our collective knowledge and skills, contributing to a more efficient development process.

5.0 Areas for Improvement

Reflecting on the project we had some areas for improvement: Although our group was on track for all our deadlines, time management did eventually become an issue. We would not account for external commitments (such as other courseworks set). This oversight led to a rush in the later stages of the project, impacting the overall thoroughness of implementation. In the future we should implement a more robust project management approach with buffer times and regular check-ins on progress, this will help accommodate unexpected delays and external commitments. Another area for improvement was evident during my development of the custom header.html without relying on Bootstrap. Opting out of Bootstrap posed several challenges; primarily, the development took longer than anticipated, and key aspects of responsive design were initially overlooked. These oversights affected not just the development timeline but also the usability of our application across different device sizes. To mitigate similar issues in future projects, I plan to incorporate more thorough preliminary research and testing phases when adopting new development methods. Early prototyping and iterative testing will be essential to identify and resolve potential problems early in the development process, ensuring that project timelines are adhered to without compromising on quality. Finally, although we maintained clear and effective communication during scheduled meetings, our interaction outside these sessions were limited, leading to occasional confusion. To mitigate this: establishing more frequent informal check-ins will be necessary.

6.0 Conclusion

The project enhanced my skills as a full-stack developer, providing a substantial learning experience in handling both backend and frontend development tasks. Our team's effective collaboration and communication allowed for us to have a successful execution, however it is important not to overlook the challenges we faced, mainly time management and (personally my) technical issues with new implementations, like the custom header. These experiences taught me valuable lessons in the importance of early testing and iteration, especially when diverging from familiar technologies. This project has prepared me for future challenges by enhancing my technical skills and by deepening my understanding of agile project management and team collaboration. These skills and insights will help me tackle future projects with greater confidence and efficiency, potentially branching out to take different roles, such as project lead or tech lead.

References and Appendix

[1] Meeting Notes from April 23rd:

Team Meeting

23 APRIL 2024 / 12:00 PM / LIBRARY 1ST FLOOR

Attendees

Mathew, Alex, Josh, Bipin, Prayush, Allia (Writer)

Agenda

Last Meeting Follow-up

1. Progress update on all outstanding coding issues

New Business

2. Inform everyone of next steps
3. Assign last week's worth of coding tasks

Notes

- Seeding complete and can be merged
- Leaderboard scrapped as it won't fulfill the streaming requirement, refocusing to the friend recommendation after advice from Stella
- Assignments as follows:
 - Alex: create issues, tools page and implementation of Mortgage Calculator API
 - Josh: create the shop feature & friend recommendation
 - Bipin: implement taxes and budgeting pathways (can be decreased to 1 if needed)
 - Prayush: testing, finish change username page styling
 - Allia: styling web page, make the FAR report available for the team to start on it
 - Other: 1st to finish their tasks will take on the final pathway coding

Action Items

1. As stated above

Next Meeting Agenda

Updates on the current progress and preparing for the Project Sponsor Meeting

[2] Meeting Notes from September 21th:

Team Meeting

21 SEPTEMBER 2024 / 10:00 AM / University of Surrey Library, Study Room 4

Attendees

Alex, Grace, Josh, Prayush, Bipin, Allia (Writer)

Agenda

Last Meeting Follow-up

1. Presenting independent research

New Business

2. Finalising the Web App's idea
3. Team member Expectations & Conflict Resolution procedures to be outlined

Notes

- **Backend research team of Alex, Josh & Bipin presented** - Josh & Bipin took lead as Alex determined his independent research to be irrelevant
 - Seneca was discussed as another inspiration alongside Duolingo
 - Possible API's were presented with the **Google Sign-In API & a Mortgage Calculator API** chosen
 - Backend languages and structure finalised: SQL Light will be used, the default User model is to be extended, admin panel will be modified to allow for new quizzes to be added
- **Frontend research team Grace, Prayush & Allia presented** - Allia was unable to complete any research due to other project related responsibilities
 - Colour palette decided and basic mockup were presented
- The teams re-split into two groups working on implementing the research to create a final design & structure
- The groups came back together and agreed on this conceptual implementation for the application
- Unable to get to the 3rd item on the agenda as only half an hour was remaining. Therefore, it pivoted to making sure that everyone was set up for success in writing their PDD sections as much as possible, with an extra meeting added into the schedule to accommodate this overrunning.

[3] Git Reference for implementation of json files and seed.py for quizzes:
csee/com2027/2023-24/Group1!22

[4] Git Reference for implementation Navbar:
csee/com2027/2023-24/Group1!18

[5] Boxicons Reference:
<https://boxicons.com/>

[6] Git Reference for implementation of header.html, footer.html, base.html, js and css:
csee/com2027/2023-24/Group1!4

[7] Git Reference for implementation of seed.py for testing:
csee/com2027/2023-24/Group1!9

[8] Problem statement proposed by Dominic Marcus Finance Consulting:

Navigating Your First Job

Create a web application to empower young adults with essential financial knowledge and skills related to their first job; pensions, payslips and budgeting. (SDG4)

Use gamification to incentivize and reward young people for their learning and engagement with healthy money habits. Solutions could incorporate understanding of investment risk.

~ Dominic Brockes



[9] From the PDD part: 8.0 Risk Analysis (10%)

8.0 Risk Analysis (10%)

In Risk Analysis, the objective is to identify, assess, and prioritise potential risks linked with the development of our web application. By comprehensively understanding these risks and implementing robust strategies to mitigate or manage them, we can significantly bolster the success and resilience of our project.

8.1 Risk Analysis Table

Below is an outline of the top 5 risks that could be identified at this time in rank order. Risks with the greatest potential to impede our progress and achievements are ranked higher. This was determined based on their Probability, which denotes their likelihood of occurrence, their Impact, which signifies the expected disruption if this were to happen, and their Severity, indicating to what

12

1 reply

Risks



Allia T
7:22 PM Mar 7

@bjgurung153@gmail.com

you think that you could get this to fit the page by cutting some stuff. Make it more of a quick summary

1 reply

[10] From the PDD part: 9.8 Risk Analysis table:

9.8 Risk Analysis

bipin rai

Table 1

Event	Probability	Impact	Severity	Rank	Action
Delay in development schedule	Low	High	High	1	Implement proactive project management techniques to monitor and track progress regularly. Break down project tasks into manageable segments and establish clear milestones and deadlines. Conduct regular progress reviews with the team to identify potential delays early on. Allocate additional resources or adjust project timelines as needed to mitigate delays. Implement contingency plans to manage unforeseen challenges or setbacks effectively. Work collaboratively to find solutions that minimise the impact on project objectives and deadlines.
Ineffective communication (between	Low	High	High	2	Encourage open communication, use collaboration tools like Teams, Google Docs, resolve conflicts proactively.

20

members)					
Difficulty in implementing APIs	Medium	Medium	Medium	3	Allocate contingency time for API integration, explore alternative APIs/ have backup APIs, if necessary.

[11] From the PDD 5.3 Individual Goals (5.2.5 Bipin):

5.3 Individual Goals

5.2.1 Allia - By week 6, my goal is to become a proficient project manager, gauged through feedback forms collected from each team member.

5.2.2 Prayush - Within the completion of the coding phase, I aim to have elevated my front-end development skills through online resources and applying them throughout the coding phase.

5.2.3 Josh - By week 6, I aim to gain experience in setting up and managing Git repositories, which will be measured by successfully managing these Git repositories.

5.2.4 Alex - Throughout the coding phase, I aim to further my back-end development skills in Python's Django framework by successfully modelling and implementing features to be managed via a database, measured by the efficiency of the database system.

5.2.5 Bipin - By week 12 to have gained proficiency in using Git for effective code management and collaboration, focusing on version control. Measuring my proficiency by successfully aiding the management of code and collaboration with team members.

5.2.6 Grace - By the project's end, to have improved my API knowledge and gained experience in GitLab for CI/CD in development cycles. I plan to assess my progress by successfully integrating APIs and utilising GitLab for CI/CD.

[12] From FAR 4.0 Individual Evaluations (section 4.6 Bipin):

4.6 Bipin

Throughout the project, I gained experience in Git for effective code management and collaboration. Using Git for the project involved exploring its functionalities, mainly the utilisation of branches. I encountered challenges with Git's trunk system, such as the risk of

merge conflicts stemming from outdated branches. These problems provided me valuable learning opportunities, with the help of my group we were able to troubleshoot effectively and understand the importance of maintaining up-to-date branches for effective collaboration.

As a Fullstack developer, my involvement in various project tasks helped me gain experience in frontend and backend development practices. For example, I had the opportunity to implement JavaScript to create the header.html. Implementing JavaScript for the header was a departure from our usual bootstrap-centric approach. This shift allowed me to explore alternative methods of frontend development and gain a deeper understanding of JavaScript's role in enhancing user interfaces. Notably, when I was creating the side header, I gained experience in developing frontend html without any pre existing backend.

[13] From the FAR section 3.0 Test Suites:

3.0 Test Suites

Each test is designed to validate specific functionalities and features of the application. While all test suites are important, we've highlighted here these particular suites due to their significance in covering key features and functionalities such as user management, quiz systems, avatar customisation and data validation.

3.1 User Management Tests

- **File Location:** app_user/tests.py
- **Purpose:** Tests that ensure user management functionalities such as registration, login, and profile management operate correctly.
- **Functionalities Covered:**
 - User Registration: Validates successful user registration with appropriate data handling for both valid and invalid input scenarios. Refer to [1] for details on the validation of form submissions for user registration.
 - User Login: Tests the login process to ensure secure and correct user access. Code Reference: [2] demonstrates how the system handles both successful and failed login attempts.
 - Profile Management: Checks the functionality that allows users to update their username, ensuring changes are saved properly. Code Reference: [3] provides the method for testing the update of user profiles.
- **Expected Outcomes:**
 - User Registration: A successful registration should create a new user account when the valid credentials are provided, confirming the system's ability to

[14] From the FAR section 3.4 Delivery:

3.4 Delivery

The project was delivered in a Docker Container as it allows for the app to be sent to the Stakeholders such that they can both run and test the application without needing to install Python and all the required modules as these were included in the container. This, therefore, meets all our requirements without the need to deploy on a host platform. Hence this was chosen as the primary and, consequently, only form of deployment for the web application.

Deployment on a host platform was considered, however this option wasn't chosen both from not meeting the module's requirements and thereby being something extra, as well as potentially impacting our cost constraint.

Furthermore, while researching possible options [1], different technical constraints such as the site needing to be static and/or having specific documentation constraints, made the one's researched into unsuitable for our web application. These constraints should have been considered at the beginning of development, rather than at the end when deployment was being considered in order to make deployment on those platforms a functional option.

Finally, all the host sites would have a learning curve to be able to properly deploy onto their site. This time sunk into learning how to deploy onto one of these platforms as well as researching the best one and altering the code to meet all of its constraints wasn't worth being able to deploy the app on it when a Docker Container met all of the requirements.

[15] From the FAR section 3.3 Evaluation:

Suites

A

Allia T

12:22 PM Apr 24

@bjgurung153@gmail.com

TODO by 4th May

Assigned to you

A

Allia T

3:24 PM Apr 24

Edit: due on the 8th May

A

Allia T

1:02 PM May 11

Marked as done

Adding a comment will re-open this discussion...

Delivery

A

Allia T

12:20 PM Apr 24

@bjgurung153@gmail.com

TODO by 3rd May

(If you would rather write up Evaluation speak to Bipin about switching. If you do however please let me know.)

Show less

A

Allia T

10:38 AM May 12

Marked as done

Adding a comment will re-open this discussion...

3.3 Evaluation

3.2.1 Project Criteria Achievement:

The project successfully met key criterias specified in the Project Definition Document (PDD). The project was built using Django, so the project is robust and scalable, allowing for easy management and deployment within the Docker container. Approval from the Project Sponsor was obtained, confirming that the project aligns with SDG4 and SDG8. APIs outlined in section 3.3 (in the PDD) were integrated into the application, with the exception of a live leaderboard, which was quickly replaced with the recommended friend and added friends features. Three pathways of financial knowledge (Pension, Taxes and Budgeting) were implemented, each enriched with interactive quizzes and informational content, meeting the initial specifications detailed in section 2.21 (in the PDD).

Gamification features such as avatars, decorations and a points system were incorporated to boost the user engagement and interaction within the application. Originally we aimed to

Evaluation

A Allia T
12:19 PM Apr 24

@bjgurung153@gmail.com
TODO by 3rd May

(If you would rather write up delivery speak to Alex about switching. If you do however please let me know.)

[Show less](#)

Assigned to you

A Allia T
6:49 PM May 12

Marked as done

[16] From the FAR section 2.1 Front-End Design

2.0 Final Design

2.1 Front-End Design

2.1.1 Design

The high level requirements for the financial education app were converted into a user-friendly and interactive design to help empower young adults with essential financial knowledge. Our stakeholder's vision, Domine Brockes from Dominic Marcus Financial Consultancy Ltd, highlighted the importance of gamification, accessibility, and user friendly navigation to help empower young adults with essential financial knowledge. In translating stakeholder requirements into design choice we initially researched similar applications which were simple but also interactive. [1]

One of our objectives was to create a UI/UX design that prioritised ease of navigation and accessibility. To help achieve our objective the selection of a suitable colour scheme was necessary. Multiple palettes were made by using Coolors.co. [2] This is because we wanted to ensure that the colour palette allows for sufficient contrasts between text and

Assigned to you

Design

A Allia T
6:56 PM May 12

@bjgurung153@gmail.com

Edit 2.1

Reply or add others with @

[17] From the FAR section 3.0 Team Evaluation

3.0 Team Evaluation

3.1 Teamwork

Effective and good quality teamwork has been demonstrated throughout the project through excellent team member conduct, consistent communication via the WhatsApp chat & direct messaging making Leads easy to reach at all times. Tasks have been delegated well due to proper planning in the concept stage, and effort has been high from all team members. [1]

This was the sentiment from the whole team who when asked, rated the groups teamwork ability on average a 9.5/10. [1]

However, communication in the weeks directly preceding and exceeding the Easter Break was lacking. This was fixed after both the Leads and Team reacclimated to University life. [1]

Assigned to you

Evaluation

A Allia T
1:00 PM May 11

@bjgurung153@gmail.com

Edit this section for 23:59 12th May

Reply or add others with @

[18] From the FAR section 1.0 How to run the application.

1.0 How to run the application

1.1 Starting & Accessing the App

The app has been provided in a Docker container with all files. To start the app, simply start the container and then navigate to `localhost:8000` in your browser. This should take you to the landing page where you will be able to navigate through the app. If the page isn't loading, check that the server has started and then reload the page. You can navigate around the website using the sidebar on the left, which can be opened or closed using the button in the top left. It is suggested that you log in first as most pages will require you to do so.

1.1.1 Standard Login

The first way to log in is to use the standard method. You can either create a new account on the signup page or use a pre-existing account. We have included a superuser for testing purposes that you can use, and to sign in simply navigate to the login page and enter the username and password as 'admin'. If you prefer to make your own account, navigate to the signup page from either the log in or landing page and then fill out all the fields. Once you click 'Register' you should be taken back to the login page where you can enter the username and password of the account you just made. After you have logged in you should be taken to the accounts page where you can view the details of your account and change settings like your username. You can also see your account details on the sidebar in the bottom left, where

